

Experiment 8

Problem:

- a) Find length of string without library function.
- b) Check if entered string is palindrome.

Student Details:

Name: madhiya zaid javed Roll No: 36 UIN: 251M040 Division: F

Code (C):

```
#include <stdio.h>
#include <string.h>
int length_no_lib(char *s){
    int i=0; while(s[i]!='\0') i++; return i;
}
int is_palindrome(char *s){
    int L = length_no_lib(s), i=0, j=L-1;
    while(i<j){ if(s[i]!=s[j]) return 0; i++; j--; }
    return 1;
}
int main(){
    char name[]="madhiya zaid javed"; int roll=36; char uin[]="251M040"; char divc='F';
    char str[100];
    printf("%s Roll:%d UIN:%s Div:%c\n",name,roll,uin,divc);
    printf("Enter a string (no spaces): ");
    if(scanf("%s", str)!=1) return 0;
    printf("Length (without library) = %d\n", length_no_lib(str));
    if(is_palindrome(str)) printf("The string is a palindrome\n");
    else printf("The string is not a palindrome\n");
    return 0;
}
```

Sample Input and Output (example run):

Input: radar

Output:

Length (without library) = 5

The string is a palindrome